

1. Administrative & Study Identification

| Field | Data Entry / Details | | :--- | :--- | | **Full Study Title** | A Study of Belzutifan (MK-6482) and Metformin in Healthy Adult Participants | | **Short Title/Acronym** | MK-6482-039 | | **Protocol Number** | MK-6482-039 | | **NCT Number** | NCT07121959 | | **Posting ID** | 989731945171074511 | | **Sponsor / Company** | Merck / Merck Sharp & Dohme LLC | | **Investigational Product** | Belzutifan (MK-6482) and Metformin | | **ATC Code (Metformin)** | A10BA02 | | **Known Trade Names (Metformin)** | Jentaduetto, PrandiMet, Actoplus Met, Janumet, Glumetza, Orabet Metformin, Riomet, Fortamet, Metforming, Avandamet, Metaglip, Glucovance, Glucamet, Glucophage, Trijardy, Qternmet, Segluromet, Synjardy, Xigduo, Kazano, Kombiglyze, Invokamet | | **Phase of Development** | Phase 1 (PHASE1) | | **Trial Status** | COMPLETED | | **Medical Monitor** | Clinical Operations Lead, Merck |

2. Study Synopsis & Rationale

2.1 Study Objective Summary/Primary Objective: The goal of this study is to learn how belzutifan (MK-6482) affects the levels of metformin in a healthy person's body over time. Researchers will study levels of metformin in the blood and urine in healthy volunteers after taking metformin alone (Period 1) and metformin with belzutifan (Period 2). **Secondary Objectives:** To evaluate the safety and tolerability of belzutifan and metformin when co-administered in healthy volunteers.

2.2 Background & Rationale Belzutifan is a HIF-2α inhibitor studied for treating certain conditions, including rare kidney tumors. Metformin is a widely utilized medication for blood sugar control in diabetes. Because patients may eventually be prescribed both therapies concurrently, this study is being conducted to understand potential pharmacokinetic drug-drug interactions. Understanding these interactions is a critical step in ensuring patient safety and determining if dosage adjustments are necessary when these drugs are used together.

3. Study Design & Methodology

3.1 Design Framework Study Type: INTERVENTIONAL (Clinical Trial) **Control Type:** Single-arm (Fixed-sequence, self-controlled cross-over) **Blinding:** Open-label **Randomization:** N/A (Fixed sequence design)

3.2 Planned Enrollment & Duration Targeted Enrollment (N): 14 healthy adult volunteers **Number of Sites:** 1 site (Celerion, Tempe, AZ, United States) **Trial Status:** COMPLETED

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy adult volunteers 2. Is a continuous non-smoker for at least 3 months prior to dosing. 3. Has body mass index (BMI) ≥ 18.0 and ≤ 42.0 kg/m^2 . 4. Is able to swallow multiple tablets.

4.2 Exclusion Criteria 1. Has a history of cancer (malignancy). 2. Has a history of anemia within the last 5 years. 3. Has a known history or presence of metabolic acidosis, ketoacidosis, or lactic acidosis. 4. Has a hemoglobin level below the lower limit of normal. 5. Has a pulse oximetry reading less than 92% at rest. 6. Is known to be a cytochrome P450 (CYP2C19) poor metabolizer or likely poor metabolizer based on genotyping prior to the screening visit or is determined to be a CYP2C19 poor metabolizer.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Period 1:** Single dose of metformin administered on Day 1. * **Period 2:** Belzutifan administered daily for 5 consecutive days, with a single dose of metformin co-administered on Day 4. **Route of Administration:** Oral (Tablets) **Duration of Treatment:** Total of 6 days of active dosing across two periods.

5.2 Concomitant & Prohibited Therapy Permitted: Standard hydration and meals as directed by the clinical site restrictions. **Prohibited:** Concomitant medications, especially those known to interfere with CYP2C19 metabolism or those that could alter the absorption or excretion of metformin/belzutifan.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	1-2 Weeks Prior Informed Consent, Genotyping, Eligibility Labs Period 1 1 Day Single dose of metformin, PK blood & urine sampling Period 2 5 Days Belzutifan daily (Days 1-5), Metformin on Day 4, PK sampling Follow-Up 4 Weeks Post-Dose Safety evaluation, monitoring for adverse events

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Area Under the Concentration Versus Time Curve From Time Zero to Infinity ($AUC_{0-\infty}$) for Metformin in Plasma: Blood samples will be collected to determine the $AUC_{0-\infty}$ of metformin in plasma.

7.2 Secondary Outcomes 1. Number of Participants Who Experience One or More Adverse Events (AEs). 2. Number of Participants Who Discontinue Study Drug Due to an AE. 3. Area Under the Concentration Versus Time Curve from 0 to the Last Quantifiable Sample (AUC_{0-last}) of Metformin.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via direct observation in the clinical unit and outpatient follow-up. Standard monitoring for events like tiredness, low red blood cell counts, and upset stomach. **Serious Adverse Events (SAEs):** Any life-threatening event or event requiring hospitalization is reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety review committee run by the study Sponsor to oversee Phase 1 safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set; Safety Set. **Sample Size Justification:** A sample size of approximately targeted enrollment ($N=14$) is adequately powered for characterizing standard intra-subject PK variations in a Phase 1 drug-drug interaction study. **Handling of Missing Data:** Standard PK modeling guidelines; concentration values below the limit of quantification are treated as zero prior to C_{max} , and missing values in terminal phases are excluded from AUC estimations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous onsite monitoring during Periods 1 and 2 confinement, followed by outpatient safety checks. **Audit Trail:** eCRF entry tracking, clinical sample chain-of-custody, and internal data clarification protocols to ensure valid PK outcomes.

11. Study Identifiers & Governance

Primary ID: {{MK-6482-039}} **Posting ID:** {{989731945171074511}}
Registry Number: {{NCT07121959}} **Sponsor/Lead Organization:** {{Merck}}
Study Title: {{A Study of Belzutifan (MK-6482) and Metformin in Healthy Adult Participants (MK-6482-039)}} **Official Regulatory Title:** {{An Open-Label, Phase 1 Study to Characterize the Effects of Belzutifan on the Pharmacokinetics of a MATE1/2K Substrate, Metformin, in Healthy Adult Participants}}

12. Clinical Framework (Pharmacokinetic Specific)

Disease Areas / Conditions: {{Healthy}} **Preferred UMLS Name:** {{Healthy Human}} **Diagnosis Threshold:** {{BMI 18.0 - 42.0 kg/m2, Normal Labs, Non-Smoker}} **Medical Methodology:** {{Fixed-sequence crossover PK profiling}} **Optimization Target:** {{Metformin AUC0-inf and Cmax characterization}}

13. Study Procedures & Methodology

Management Pathway: {{Phase 1 fixed-sequence clinic admission pathway}} **Education Protocol (HCP):** {{Rigorous PK sampling timelines and dosing administration protocol}} **Education Protocol (Patient):** {{Clinic dietary restrictions, smoking cessation rules, AE reporting}} **Quality Audit Mechanism:** {{Strict PK sample handling & clinical site protocol adherence monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{Approx. 6 weeks (including screening & follow-up)}} **Onsite Visit Cadence:** {{In-patient confinement during Period 1 (1 day) and Period 2 (5 days)}} **Remote Monitoring Frequency:** {{Post-treatment AE check-ins}} **Checklist Review Interval:** {{4-week post-treatment safety review}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				